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FINAL REPORT

WHEN AI DISAGREES WITH YOU: HOW UNIVERSITY STUDENTS RESPOND TO UNCERTAIN AI ANSWERS

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Abstract

As generative artificial intelligence (AI) assistants become widespread in university coursework, students routinely encounter uncertain or questionable outputs—such as an AI contradicting a student's own reasoning, disagreements between separate AI tools, unverifiable bibliographic citations, unclear intermediate reasoning steps, or answer reversals under simple questioning. Understanding how students react initially to such uncertainty is essential for assessing critical AI literacy in higher education. Rather than imposing predefined categories or grading student actions as right or wrong, this project seeks to uncover whether natural behavioral patterns emerge in students' immediate responses across realistic coursework conflict scenarios. Specifically, we aim to identify distinct response strategies, examine how different agglomerative hierarchical clustering linkage methods behave on nominal survey data, and evaluate whether the resulting groupings are stable and associated with students' academic seniority and frequency of AI use. To investigate these objectives, we designed and administered an anonymous scenario-based survey at the Vietnamese-German University (VGU), recording each respondent's immediate action choice across multiple AI-uncertainty scenarios alongside withheld validation variables for year of study and weekly AI usage frequency. After cleaning and harmonizing the nominal survey data, we measured pairwise student dissimilarities using the Normalized Hamming (Gower) metric to preserve the unordered categorical response space without arbitrary numerical encoding. We then constructed and compared hierarchical clusterings using Single Linkage, Complete Linkage, and Average Linkage (UPGMA), evaluated dendrogram quality and cluster separation through cophenetic correlation and silhouette analysis, tested partition robustness via row-order permutations and random subsampling with the Adjusted Rand Index, profiled the resulting behavioral clusters, and performed external statistical validation against year of study and weekly AI usage.

Keywords: Multivariate Statistics, Agglomerative Hierarchical Clustering, Normalized Hamming Distance, Gower Dissimilarity, UPGMA, Silhouette Analysis, Adjusted Rand Index, AI Literacy.

1 Introduction, Motivation, and Research Questions

Generative AI tools such as ChatGPT, Gemini, and Claude are now a normal part of university coursework. While these tools can help students solve problems faster, they are not always reliable. Students often see AI answers that sound convincing but are wrong, include fake paper references, skip or hide math steps, or suddenly change their answer when asked a simple "Are you sure?". What students choose to do first when they face an unclear or conflicting AI answer decides whether AI helps them learn better or leads them to accept wrong information.

Instead of forcing students into fixed categories beforehand or grading their choices as "right" or "wrong," our goal is to see whether natural groups of student behavior appear directly from their survey answers. Based on this goal, we focus on three main research questions:

- **RQ1 (Behavioral Groups):** When an AI answer is confusing or disagrees with other information across coursework scenarios (Q_1–Q_5), do university students show clear patterns in what they choose to do first?
- **RQ2 (Linkage Sensitivity):** How do the resulting cluster groups change when we use Single Linkage, Complete Linkage, and Average Linkage (UPGMA) on a discrete Hamming distance space?
- **RQ3 (Partition Stability & External Validity):** Do the student clusters stay stable when we shuffle the row order or take 80% random subsamples, and do they differ across external variables not used in clustering—specifically Year of Study (S_4) and Weekly AI Usage Frequency (Q_6)?

2 Theoretical Framework and Method Explanation

2.1 General Objectives of Hierarchical Cluster Analysis

Following the course formulation of multivariate cluster analysis [1], cluster analysis is an unsupervised exploratory technique for partitioning a multivariate dataset of n sample units measured on p variables into "natural" homogeneous groups. We assume the n observed sample units originate from an unknown number of subpopulations with no *a priori* group labels. Let

$$X_{ik} = \text{response for variable } k \in \{1, \ldots, p\} \text{ in sample unit } i \in \{1, \ldots, n\}, \tag{1}$$

so that each respondent i is represented by a p-dimensional observation vector $\vec{X}_i = (X_{i1}, X_{i2}, \ldots, X_{ip})^\top$.

Cluster analysis methods divide broadly into *non-hierarchical methods* (such as MacQueen's K-means algorithm, which iteratively reallocates observations to K pre-specified centroids) and *hierarchical methods*. Hierarchical methods construct a nested sequence of partitions represented visually as a two-dimensional tree diagram (*dendrogram*), operating in one of two directions:

1. **Agglomerative Hierarchical Algorithms (Bottom-Up):** Start at step 0 with n singleton clusters $\{C_1, \ldots, C_n\}$ where $C_i = \{\vec{X}_i\}$. At each subsequent step $t = 1, \ldots, n-1$, the two closest existing clusters C_r and C_s (according to an inter-cluster linkage dissimilarity d_{rs}) are merged into a single cluster $C_r \cup C_s$, reducing the number of clusters by one until all n sample units form a single cluster of size n.
2. **Divisive Hierarchical Algorithms (Top-Down):** Start with all n observations in a single cluster and recursively split one existing cluster into two daughter clusters at each step. Because divisive splitting requires evaluating $2^{n-1} - 1$ possible bipartitions at the first step alone, agglomerative algorithms are used far more widely in practice.

2.2 Nominal Feature Space and Normalized Hamming Distance

To cluster sample units, we first require a pairwise *measure of association* (dissimilarity) $d(\vec{X}_i, \vec{X}_j)$ between observation vectors $\vec{X}_i$ and $\vec{X}_j$. As established in lecture [1], a valid metric distance function must satisfy four axiomatic properties for all $\vec{X}_i, \vec{X}_j, \vec{X}_k$:

$$\textbf{1. Symmetry:} \quad d(\vec{X}_i, \vec{X}_j) = d(\vec{X}_j, \vec{X}_i) \tag{2}$$

$$\textbf{2. Positivity:} \quad d(\vec{X}_i, \vec{X}_j) > 0 \quad \text{if } \vec{X}_i \neq \vec{X}_j \tag{3}$$

$$\textbf{3. Identity:} \quad d(\vec{X}_i, \vec{X}_j) = 0 \quad \text{if } \vec{X}_i = \vec{X}_j \tag{4}$$

$$\textbf{4. Triangle Inequality:} \quad d(\vec{X}_i, \vec{X}_k) \leq d(\vec{X}_i, \vec{X}_j) + d(\vec{X}_j, \vec{X}_k) \tag{5}$$

Standard continuous distance metrics—such as Euclidean distance $d(\vec{X}_i, \vec{X}_j) = \sqrt{\sum_{k=1}^{p}(X_{ik} - X_{jk})^2}$, Minkowski distance $\left[\sum_{k=1}^{p} |X_{ik} - X_{jk}|^m\right]^{1/m}$, and the Canberra metric—require arithmetic differences $X_{ik} - X_{jk}$. However, in our survey, each variable $X_{ik} \in \mathcal{A}_k$ represents an *unordered nominal category* (e.g., Option `A` = checking course materials, Option `C` = asking the same AI, Option `D` = asking a different AI or classmate). Encoding $\{\texttt{A}, \texttt{B}, \texttt{C}, \texttt{D}, \texttt{E}, \texttt{F}\}$ as integers $\{1, 2, 3, 4, 5, 6\}$ and computing Euclidean distances would impose an artificial linear ordering where Option `A` is five times farther from `F` than from `B`.

Instead, we define dissimilarity on the nominal space $\mathcal{X} = \mathcal{A}_1 \times \cdots \times \mathcal{A}_p$ using the **Normalized Hamming Distance** (which coincides exactly with **Gower's Dissimilarity Coefficient** [2] for equally weighted qualitative variables with complete data):

$$d_H(\vec{X}_i, \vec{X}_j) = \frac{1}{p} \sum_{k=1}^{p} \delta(X_{ik}, X_{jk}), \qquad \text{where } \delta(X_{ik}, X_{jk}) = \begin{cases} 0 & \text{if } X_{ik} = X_{jk}, \\ 1 & \text{if } X_{ik} \neq X_{jk}. \end{cases} \tag{6}$$

For $p = 5$ scenarios, $d_H(\vec{X}_i, \vec{X}_j)$ measures the exact proportion of scenarios on which student i and student j chose different actions, taking values in the discrete six-point lattice

$$d_H(\vec{X}_i, \vec{X}_j) \in \{0.0,\ 0.2,\ 0.4,\ 0.6,\ 0.8,\ 1.0\}\,. \tag{7}$$

It is straightforward to verify that d_H satisfies all four metric axioms (2)–(5) on the space of response profiles: $\delta(a, b) = \delta(b, a)$ ensures symmetry; $\delta(a, b) \geq 0$ with equality if and only if $a = b$ ensures positivity and identity on distinct profiles $\vec{X}_i \neq \vec{X}_j$ (while two distinct students sharing an identical 5-question profile $\vec{X}_i = \vec{X}_j$ legitimately have $d_H = 0$); and since $\delta(X_{ik}, X_{kk'}) \leq \delta(X_{ik}, X_{jk}) + \delta(X_{jk}, X_{kk'})$ holds coordinate-wise for the discrete metric, summing over $k = 1, \ldots, p$ and dividing by p preserves the triangle inequality.

2.3 Mathematical Formulation of Linkage Criteria

Suppose cluster 1 contains k observation vectors $\{\vec{X}_1, \vec{X}_2, \ldots, \vec{X}_k\}$ and cluster 2 contains l observation vectors $\{\vec{Y}_1, \vec{Y}_2, \ldots, \vec{Y}_l\}$. Following the notation of Prof. Dr. C. Andersson [1], the five standard agglomerative linkage methods define inter-cluster dissimilarity d_{12} as follows:

1. **Single Linkage (Nearest-Neighbor):** Defines inter-cluster distance as the minimum distance between any single observation in cluster 1 and any single observation in cluster 2:

$$d_{12}^{\text{single}} = \min_{1 \leq i \leq k,\, 1 \leq j \leq l} d(\vec{X}_i, \vec{Y}_j). \tag{8}$$

   At each stage, the two clusters with the smallest d_{12}^{single} are merged. While sensitive to local bridges, Single Linkage suffers from the *chaining phenomenon*: two large, otherwise distant clusters are merged prematurely whenever a chain of intermediate points connects them.

2. **Complete Linkage (Furthest-Neighbor):** Defines inter-cluster distance as the maximum distance between any single observation in cluster 1 and any single observation in cluster 2:

$$d_{12}^{\text{complete}} = \max_{1\le i\le k,\, 1\le j\le l} d(\vec{X}_i, \vec{Y}_j). \tag{9}$$

Complete linkage enforces compact, spherical-diameter clusters because merging two groups requires every pair across the combined group to lie within d_{12}^{complete}.

3. **Average Linkage (UPGMA — Unweighted Pair Group Mean Averaging):** Defines inter-cluster distance as the arithmetic average of all $k \times l$ pairwise distances between observations in cluster 1 and observations in cluster 2:

$$d_{12}^{\text{average}} = \frac{1}{kl} \sum_{i=1}^{k} \sum_{j=1}^{l} d(\vec{X}_i, \vec{Y}_j). \tag{10}$$

Average linkage provides a compromise between the chaining tendency of Single Linkage and the extreme outlier sensitivity of Complete Linkage, serving as our primary clustering method.

4. **Centroid Method and Ward's Minimum Variance Method (Why Not Applied Directly):** In the **Centroid Method**, inter-cluster distance is the distance between cluster mean vectors, $d_{12}^{\text{centroid}} = d(\bar{\vec{x}}_1, \bar{\vec{y}}_2)$. In **Ward's Method**, clustering is formulated as an ANOVA-based variance minimization problem [1], merging the pair of clusters that minimizes the increase in the combined Error Sum of Squares (SSE) or equivalently maximizes R^2:

$$SSE = \sum_i \sum_j \sum_{k=1}^{p} |X_{ijk} - \bar{x}_{i\cdot k}|^2, \qquad SST = \sum_i \sum_j \sum_{k=1}^{p} |X_{ijk} - \bar{x}_{\cdot\cdot k}|^2, \qquad R^2 = \frac{SST - SSE}{SST}. \tag{11}$$

Because X_{ijk} in our study are unordered nominal labels (`A`–`F`), sample means $\bar{x}_{i\cdot k}$ and squared deviations $|X_{ijk} - \bar{x}_{i\cdot k}|^2$ are undefined without arbitrary numerical coding or high-dimensional one-hot dummy expansion. In contrast, Single, Complete, and Average Linkage operate strictly on the pairwise dissimilarity matrix $D = [d_H(\vec{X}_i, \vec{X}_j)]_{n\times n}$, respecting the nominal scale of the survey.

2.4 Strengths and Weaknesses of the Methodology

Strengths: (1) Agglomerative clustering with Normalized Hamming distance requires no parametric distributional assumptions or arbitrary numerical scoring of categorical choices; (2) it reveals the full multi-scale hierarchy of student behavior (e.g., showing how broad self-verification splits into textbook lookup versus independent derivation); and (3) Average Linkage smoothing transforms discrete pairwise distances into fine-grained rational cluster distances.

Weaknesses and Theoretical Nuances:

- *Equal-Distance Assumption Within Questions:* Normalized Hamming distance treats any mismatch $\delta(X_{ik}, X_{jk}) = 1$ equally,even if two options (such as checking lecture notes vs. solving independently) are conceptually closer than accepting AI uncritically.

- *Discrete Distance Lattice & Tie-Breaking Sensitivity:* With $p = 5$ questions, pairwise distances take only 6 values. Under Single and Complete Linkage, since min and max are order statistics of $\{0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, *every merge height in the entire dendrogram is restricted to those 6 numbers.* Thousands of tied pairs force the algorithm to break ties according to respondent row order in the dataset. Average Linkage (10) overcomes this limitation because dividing by kl produces rational merge heights with far fewer ties.

- *Varying Category Counts (5 vs. 6 Options):* Scenarios Q_1, Q_2, Q_5 have $M_k = 6$ answer options (`A`–`F`), whereas Q_3, Q_4 have $M_k = 5$ options (`A`–`E`). While each question receives an equal

weight of $1/5 = 0.20$ in Equation (6), the probability of two respondents matching purely by chance under uniform selection is $\frac{1}{5} = 20.0\%$ on Q_3, Q_4 compared to $\frac{1}{6} \approx 16.7\%$ on Q_1, Q_2, Q_5 (and empirically $\sum_m \hat{p}_{km}^2$). Thus, 6-option questions contribute slightly more to total pairwise dissimilarity on average than 5-option questions.

3 Practical Empirical Study: Survey Design and Data Collection

3.1 Survey Design and the Five AI Uncertainty Scenarios

We developed an anonymous online questionnaire via Google Forms titled *"When AI Disagrees With You: How University Students Respond to Uncertain AI Answers."* Following two screening questions verifying university enrollment (S_1) and voluntary informed consent (S_2), along with demographic items capturing study program (S_3) and Year of Study (S_4), respondents were presented with $p = 5$ realistic coursework scenarios (Q_1–Q_5) and asked to select the **single action they would most likely take first**:

- Q_1 **(AI Disagrees With Your Answer — 6 options):** You solve a multiple-choice question and choose B; AI chooses D with an explanation. (`A`: Check course material; `B`: Re-solve independently; `C`: Ask same AI why B is wrong; `D`: Ask a different AI; `E`: Change to D without checking; `F`: Keep B without checking).
- Q_2 **(Two AI Assistants Disagree — 6 options):** Two AIs give plausible explanations for B vs. D on an unsolved problem. (`A`: Check course material; `B`: Solve independently; `C`: Send 2nd explanation to 1st AI to compare; `D`: Ask a 3rd AI; `E`: Choose preferred AI without checking; `F`: Leave unanswered and move on).
- Q_3 **(A Cited Paper Cannot Be Located — 5 options):** AI cites a paper title that yields no result on Google Scholar. (`A`: Independently search another source to verify claim; `B`: Ask same AI for DOI/link; `C`: Ask different AI if claim is correct; `D`: Use claim but omit citation; `E`: Remove claim without investigating).
- Q_4 **(One Reasoning Step Is Unclear — 5 options):** You cannot see why step 3 leads to step 4 in an AI solution. (`A`: Look up concept/formula in course materials; `B`: Derive step 4 from step 3 yourself; `C`: Ask same AI to explain transition; `D`: Ask a classmate; `E`: Use solution without clarifying).
- Q_5 **(AI Changes Answer Without New Evidence — 6 options):** AI chooses B; you ask "Are you sure?"; AI apologizes and switches to D. (`A`: Check course material; `B`: Solve independently; `C`: Ask AI to compare both and explain change; `D`: Ask different AI; `E`: Accept D without checking; `F`: Keep B without checking).

Finally, respondents answered Q_6: *"On how many of the past seven days did you use a generative AI assistant for coursework?"* (0 to 7 days). Importantly, S_4 and Q_6 were withheld from the clustering algorithm and reserved strictly for external validation.

3.2 Data Preprocessing and Sample Demographics

A total of $N = 263$ responses were collected. All 263 respondents satisfied $S_1 =$ Yes and $S_2 =$ Yes with zero missing values across Q_1–Q_6. During preprocessing in `R`, respondents were assigned anonymous IDs (`R001`–`R263`). Inspection of unique text strings revealed that Option `D` in Q_1 appeared under two slight wording variants between the pilot and final form (*"Ask a different AI the same question again."* [$n = 29$] and *"Ask a different AI the original question."* [$n = 13$]); these were harmonized into a single factor level `D` ($n = 42$). Across the $6 \times 6 \times 5 \times 5 \times 6 = 5{,}400$ theoretical response combinations, the 263 students exhibited 217 distinct 5-question profiles, with the most frequent

identical profiles being `B-B-A-B-B` ($n = 8$), `D-D-C-D-D` ($n = 7$), `C-C-B-C-C` ($n = 6$), `F-F-E-D-F` ($n = 5$), and `A-A-A-A-A` ($n = 4$).

By Year of Study (S_4), the sample comprises 63 1st-year (24.0%), 66 2nd-year (25.1%), 48 3rd-year (18.3%), 68 4th-year (25.9%), and 12 Graduate/Master's students (4.6%), with 6 unspecified (2.3%). By Major (S_3), Computer Science and Engineering (CSE) accounts for 52.1% ($n = 137$), followed by ECE (3.8%), BCE (2.7%), MEN (2.7%), MEC (2.3%), BBA (1.9%), BFA (1.9%), other VGU programs (2.7%), and unspecified (30.0%). For weekly AI usage (Q_6), students reported a mean of 4.10 ± 2.12 days/week (median $= 4.0$, IQR $= 3.0$).

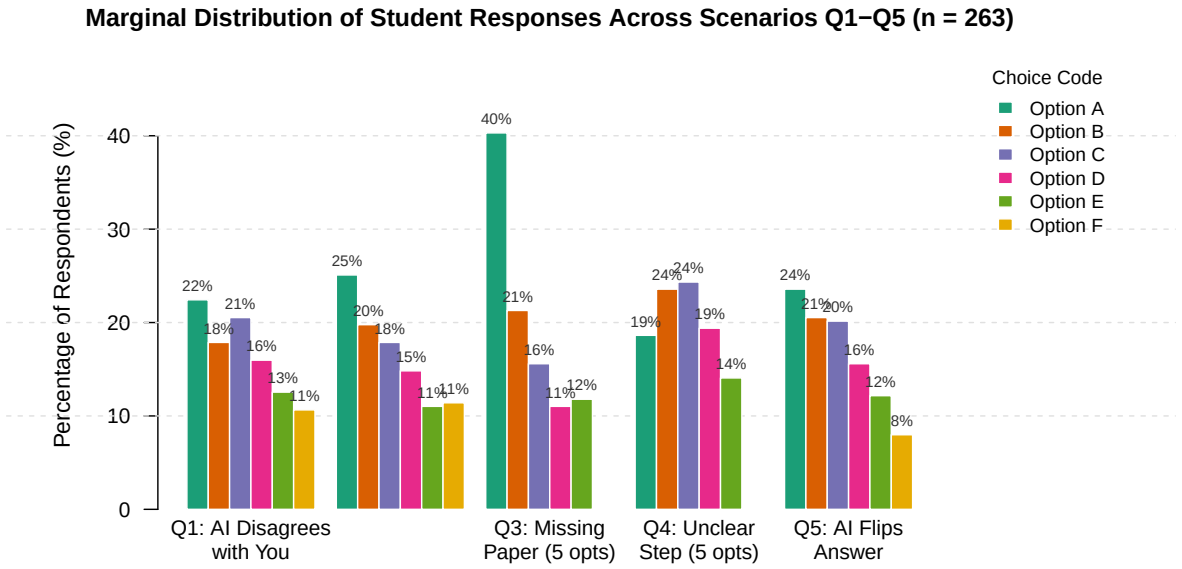


Figure 1: Marginal percentage distributions of student responses across scenarios Q_1–Q_5 ($n = 263$).

Figure 1 displays the marginal distributions across Q_1–Q_5. Checking course materials (`A`, 18.6%–25.1%), solving/deriving independently (`B`, 17.9%–23.6%), and probing the same AI (`C`, 17.9%–24.3%) are balanced across Q_1, Q_2, Q_4, Q_5. In Q_3 (missing paper), Option `A` (*independently search for another source*) reaches 40.3% ($n = 106$) because it combines textbook/library verification and independent literature search into a single 5-option item. Consequently, the empirical pairwise match rate $\sum_m \hat{p}_{km}^2$ is highest for Q_3 (25.81%) and Q_4 (20.69%) compared to the 6-option scenarios Q_1 (17.70%), Q_2 (18.12%), and Q_5 (18.38%). Across all $\binom{263}{2} = 34{,}453$ respondent pairs (Figure 2), the mean Normalized Hamming distance is 0.8016 ± 0.2005 (median $= 0.80$), with 104 pairs (0.30%) at $d_H = 0.0$, 559 (1.62%) at 0.2, 2,184 (6.34%) at 0.4, 6,220 (18.05%) at 0.6, 12,422 (36.05%) at 0.8, and 12,964 (37.63%) at 1.0.

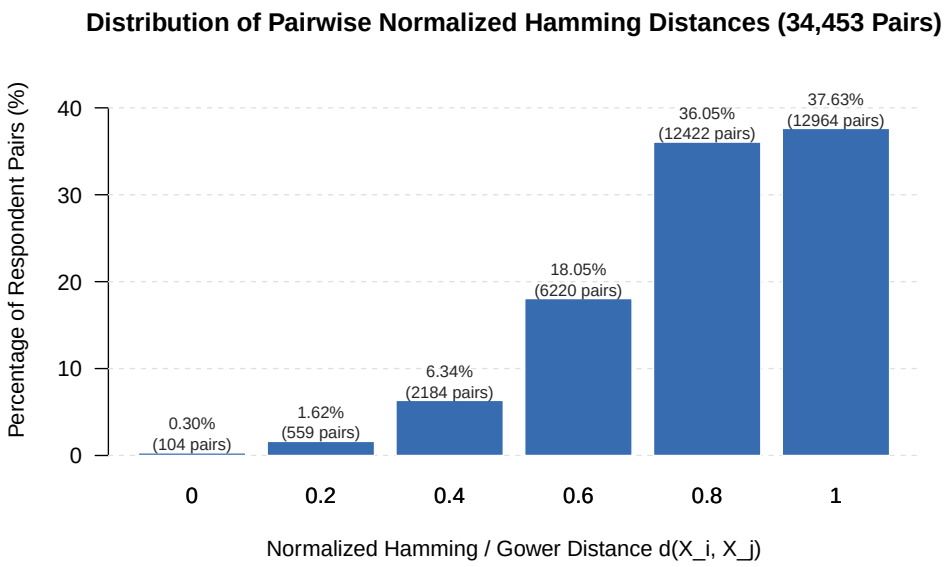


Figure 2: Empirical distribution of Normalized Hamming distances across all 34,453 pairs of respondents.

4 Empirical Results: Linkage Comparison and Cluster Profiling

4.1 Comparison of Single, Complete, and Average Linkage (RQ2)

We constructed agglomerative dendrograms from the 263×263 Gower/Hamming matrix using `stats::hclust` under Single, Complete, and Average Linkage (Figure 3). To quantify how faithfully each dendrogram preserves the original pairwise distances $d_H(\vec{X}_i, \vec{X}_j)$, we computed the **Cophenetic Correlation Coefficient** $r_{\text{coph}} = \text{Cor}(d_H, d_{\text{coph}})$, where $d_{\text{coph}}(i,j)$ is the dendrogram height at which respondents i and j first merge:

$$r_{\text{coph}}^{\text{Average}} = \mathbf{0.6275} \quad > \quad r_{\text{coph}}^{\text{Complete}} = \mathbf{0.5253} \quad \gg \quad r_{\text{coph}}^{\text{Single}} = \mathbf{0.1449}. \tag{12}$$

Average Linkage preserves the pairwise geometry substantially better than both alternatives. Examining the 262 merge heights reveals why Single and Complete Linkage struggle on discrete Hamming spaces:

- **Single Linkage Chaining:** Because $d_{12}^{\text{single}} = \min d(\vec{X}_i, \vec{Y}_j) \in \{0, 0.2, 0.4, 0.6\}$, out of 262 merges, 46 occur at $h = 0.0$ (identical profiles), 154 occur at $h = 0.2$ (differing on 1 question), 61 occur at $h = 0.4$, and 1 occurs at $h = 0.6$. Because almost every student shares at least 3 or 4 answers with *some* neighbor, Single Linkage chains 97.3%–99.6% of the entire student body into one massive cluster while peeling off singletons (258/1/1/1/1/1 at $K = 6$).
- **Complete Linkage Ceiling Degeneracy:** Because $d_{12}^{\text{complete}} = \max d(\vec{X}_i, \vec{Y}_j)$, merge heights are restricted to $\{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, and the final 15 merges all occur at the maximum ceiling height $h = 1.0$. Consequently, cutting the Complete Linkage tree at any $K \leq 16$ via `cutree` slices inside 15 tied merges at $h = 1.0$, making the partition heavily dependent on input row order.
- **Average Linkage (UPGMA) Resolution:** By averaging over cluster cardinalities $k \times l$, Average Linkage produces 66 distinct merge heights, with the top merges occurring at strictly increasing heights (0.6714, 0.6905, 0.7065, 0.7229, 0.7661, 0.7697, 0.7729, 0.7986, 0.8050, 0.8374, 0.8576, 0.8653, and 0.8732), completely eliminating tie degeneracy in the upper tree.

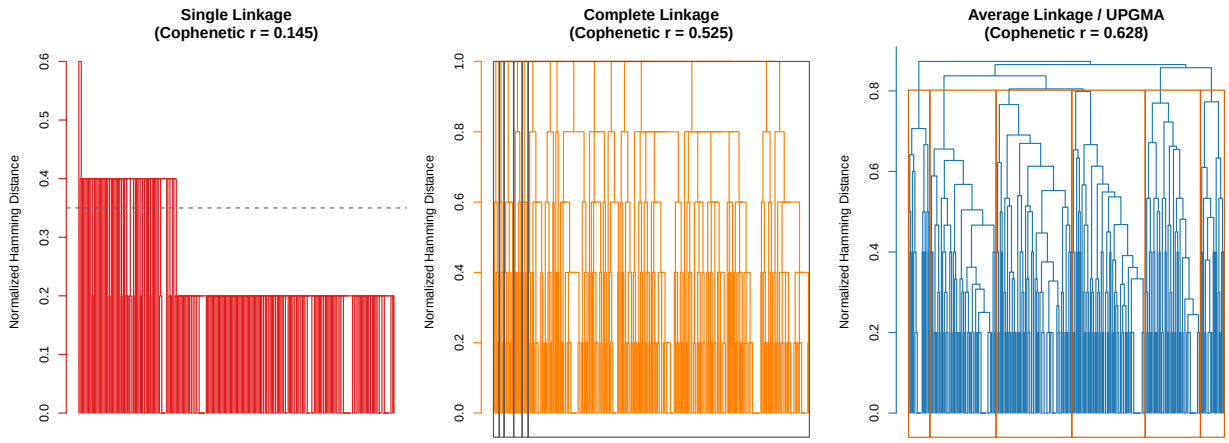


Figure 3: Comparison of Single, Complete, and Average Linkage dendrograms ($n = 263$) with $K = 6$ cut boxes.

4.2 Determining the Optimal Number of Clusters and Stability Analysis (RQ3)

Table 1 and Figure 4 summarize partition quality across $K \in \{2, \ldots, 8\}$ using the Average Silhouette Width $\bar{s} = \frac{1}{n}\sum_{i=1}^{n} \frac{b(i)-a(i)}{\max\{a(i),b(i)\}}$, negative silhouette percentage, and cluster size distribution.

Table 1: Comparison of Silhouette Widths, Cluster Sizes, and Stability (ARI) Across Linkage Methods.

Method	K	Avg Sil ($\bar{s}$)	Neg Sil (%)	Max Clust (%)	Cluster Sizes ($n_1/\ldots/n_K$)	Perm ARI	Sub80 ARI
Single	3	−0.0969	65.4%	99.2%	261/1/1	0.493 ± 0.014	0.657 ± 0.279
Single	5	−0.1778	78.7%	98.5%	259/1/1/1/1	0.292 ± 0.112	0.777 ± 0.178
Single	6	−0.2113	83.3%	98.1%	258/1/1/1/1/1	0.251 ± 0.120	0.829 ± 0.119
Complete	3	−0.0371	51.3%	96.6%	254/5/4	0.074 ± 0.144	0.120 ± 0.198
Complete	5	−0.0682	63.5%	90.9%	239/8/7/5/4	0.134 ± 0.141	0.322 ± 0.168
Complete	6	−0.0810	67.3%	89.0%	234/8/7/5/5/4	0.137 ± 0.109	0.298 ± 0.141
Average	2	+0.1030	16.7%	93.2%	245/18	0.369 ± 0.268	0.294 ± 0.260
Average	3	+0.1178	17.1%	68.1%	179/66/18	0.504 ± 0.159	0.507 ± 0.179
Average	4	+0.1186	21.7%	68.1%	179/46/20/18	0.514 ± 0.194	0.477 ± 0.208
Average	**5**	**+0.1652**	18.3%	47.1%	124/55/46/20/18	$\mathbf{0.573 \pm 0.103}$	$\mathbf{0.555 \pm 0.131}$
Average	**6**	**+0.2001**	**15.6%**	**24.0%**	63/61/55/46/20/18	$\mathbf{0.511 \pm 0.093}$	$\mathbf{0.522 \pm 0.096}$
Average	7	+0.1993	15.2%	24.0%	63/55/53/46/20/18/8	—	—
Average	8	+0.1985	17.5%	24.0%	63/55/53/46/18/12/8/8	—	—

Under Average Linkage, $\bar{s}$ rises monotonically from $K = 2$ (0.1030) to a global maximum at $\mathbf{K = 6}$ ($\bar{\mathbf{s}} = \mathbf{0.2001}$, with the lowest proportion of negative silhouettes at 15.6%), before plateauing at $K = 7, 8$ where small splinter clusters of size 8 appear. Furthermore, at $K = 6$ the largest cluster share drops from 47.1% ($n = 124$ at $K = 5$) to a well-balanced 24.0% ($n = 63$), yielding six substantive groups of sizes $63, 61, 55, 46, 20, 18$.

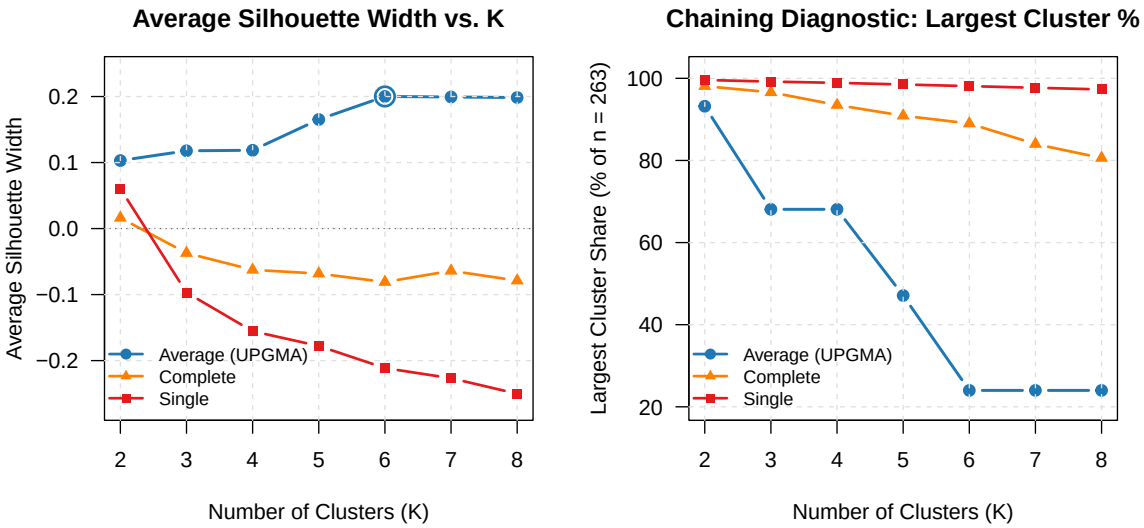


Figure 4: Left: Average Silhouette Width $\bar{s}$ across $K = 2 \ldots 8$. Right: Largest cluster percentage showing severe chaining in Single Linkage vs. balanced partitioning in Average Linkage.

To answer **RQ3**, we conducted two rigorous stability experiments (Figure 5):

1. **Row-Order Permutation Sensitivity (100 Random Shuffles):** Because internal merges at $h \in \{0.0, 0.2, 0.4\}$ contain ties, randomly permuting respondent row order tests whether tie-breaking alters final cluster membership. Complete Linkage collapses under row shuffling (mean ARI = 0.074–0.137), whereas Average Linkage achieves substantially higher order stability, peaking at $K = 5$ (ARI $= 0.573 \pm 0.103$) and $K = 6$ (ARI $= 0.511 \pm 0.093$).

2. 80% **Subsampling Stability (100 Random Subsets of $n_{\text{sub}} = 210$ Without Replacement):** Comparing each subsample partition to the full-data partition via the Adjusted Rand Index confirms strong structural recovery for Average Linkage at $K = 5$ (mean ARI $= 0.555$, median $= 0.574$, IQR $= 0.169$) and $K = 6$ (mean ARI $= 0.522$, median $= 0.532$, IQR $= 0.108$).

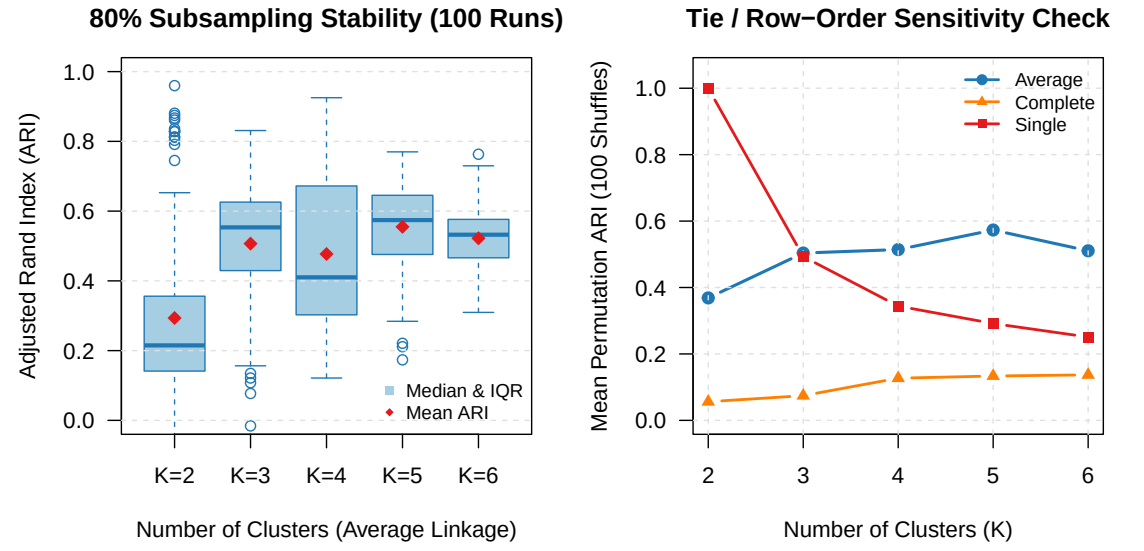


Figure 5: Left: Distribution of Adjusted Rand Index (ARI) across 100 random 80% subsamples for Average Linkage. Right: Mean row-permutation ARI across 100 random dataset shuffles.

4.3 Behavioral Archetypes and Cluster Profiling (RQ1)

Table 2 and Figure 6 present the within-cluster response percentages across all options of Q_1–Q_5 for the optimal $K = 6$ Average Linkage partition. Notably, the hierarchy between $K = 5$ and $K = 6$ has a transparent substantive interpretation: at $K = 5$, Clusters 3 and 4 form a combined 124-student group of *Self-Reliant & Course-Material Verifiers* who overwhelmingly select Option `A` in Q_3 (*independently search for another source to verify the missing paper*). At $K = 6$, this super-cluster splits cleanly into **Cluster 3 (*Independent Problem-Solvers*, $n = 61$)** and **Cluster 4 (*Course-Material Verifiers*, $n = 63$)**.

Table 2: Within-Cluster Option Percentages (%) Across Scenarios Q_1–Q_5 for Average Linkage ($K = 6$). Bold entries denote the modal response exceeding 40% within each cluster and scenario.

Cluster & Behavioral Archetype	Scenario	Opt A	Opt B	Opt C	Opt D	Opt E	Opt F	Dominant Action
C1: Cross-AI & Peer	Q_1 (Disagree)	15.2	4.3	13.0	**60.9**	6.5	0.0	Ask a different AI
Triangulators	Q_2 (2 AIs)	6.5	8.7	13.0	**47.8**	15.2	8.7	Ask a third AI
($n = 46, 17.5\%$)	Q_3 (No Paper)	15.2	4.3	**65.2**	10.9	4.3	—	Ask different AI if true
$\bar{Q}_6 = 3.67$ d/wk	Q_4 (Unclear Step)	15.2	4.3	15.2	**52.2**	13.0	—	Ask a classmate
	Q_5 (AI Flips)	8.7	10.9	6.5	**58.7**	8.7	6.5	Ask a different AI
C2: Same-AI	Q_1 (Disagree)	9.1	3.6	**65.5**	9.1	3.6	9.1	Ask same AI why B wrong
Conversationalists	Q_2 (2 AIs)	32.7	9.1	**45.5**	3.6	3.6	5.5	Send 2nd to 1st AI
($n = 55, 20.9\%$)	Q_3 (No Paper)	25.5	**65.5**	3.6	3.6	1.8	—	Ask same AI for DOI
$\bar{Q}_6 = 4.67$ d/wk	Q_4 (Unclear Step)	1.8	7.3	**83.6**	3.6	3.6	—	Ask same AI to explain
	Q_5 (AI Flips)	20.0	9.1	**56.4**	7.3	5.5	1.8	Ask AI to compare both
C3: Independent	Q_1 (Disagree)	18.0	**57.4**	8.2	3.3	8.2	4.9	Re-solve independently
Problem-Solvers	Q_2 (2 AIs)	9.8	**54.1**	8.2	9.8	6.6	11.5	Solve independently
($n = 61, 23.2\%$)	Q_3 (No Paper)	**60.7**	18.0	6.6	3.3	11.5	—	Search another source
$\bar{Q}_6 = 3.89$ d/wk	Q_4 (Unclear Step)	9.8	**70.5**	3.3	3.3	13.1	—	Derive step 4 myself
	Q_5 (AI Flips)	6.6	**62.3**	13.1	3.3	14.8	0.0	Solve independently
C4: Course-Material	Q_1 (Disagree)	**57.1**	7.9	7.9	11.1	7.9	7.9	Check course material
Verifiers	Q_2 (2 AIs)	**54.0**	0.0	9.5	11.1	14.3	11.1	Check course material
($n = 63, 24.0\%$)	Q_3 (No Paper)	**68.3**	1.6	6.3	11.1	12.7	—	Search another source
$\bar{Q}_6 = 4.40$ d/wk	Q_4 (Unclear Step)	**47.6**	14.3	7.9	20.6	9.5	—	Look up in course mat.
	Q_5 (AI Flips)	**65.1**	4.8	14.3	1.6	7.9	6.3	Check course material
C5: Uncritical AI	Q_1 (Disagree)	0.0	0.0	10.0	0.0	**85.0**	5.0	Change to D unverified
Conformers	Q_2 (2 AIs)	10.0	**40.0**	10.0	0.0	35.0	5.0	Solve / Pick usual AI
($n = 20, 7.6\%$)	Q_3 (No Paper)	15.0	10.0	5.0	**60.0**	10.0	—	Use claim, omit citation
$\bar{Q}_6 = 2.85$ d/wk	Q_4 (Unclear Step)	25.0	15.0	10.0	10.0	**40.0**	—	Use unclarified solution
	Q_5 (AI Flips)	10.0	10.0	0.0	30.0	**45.0**	5.0	Accept D unverified
C6: AI-Skeptical	Q_1 (Disagree)	0.0	16.7	0.0	0.0	5.6	**77.8**	Keep own B unverified
Anchors	Q_2 (2 AIs)	16.7	11.1	16.7	11.1	0.0	**44.4**	Leave task unanswered
($n = 18, 6.8\%$)	Q_3 (No Paper)	11.1	22.2	0.0	5.6	**61.1**	—	Remove claim immediately
$\bar{Q}_6 = 4.44$ d/wk	Q_4 (Unclear Step)	0.0	5.6	11.1	**44.4**	38.9	—	Ask classmate / Use as-is
	Q_5 (AI Flips)	0.0	5.6	11.1	5.6	11.1	**66.7**	Keep B without checking

The six clusters represent highly coherent behavioral archetypes:

1. **C4 — Course-Material Verifiers ($n = 63, 24.0\%$):** The largest group anchors verification in authoritative academic sources, checking course materials in Q_1 (57.1%), Q_2 (54.0%), Q_4 (47.6%), and Q_5 (65.1%), and searching independent scholarly literature in Q_3 (68.3%).

2. **C3 — Independent Problem-Solvers ($n = 61, 23.2\%$):** Rather than looking up reference answers, these students rely on their own analytical reasoning—re-solving problems independently in Q_1 (57.4%), Q_2 (54.1%), and Q_5 (62.3%), deriving the missing step themselves in Q_4 (70.5%), and independently verifying literature in Q_3 (60.7%).

3. **C2 — Same-AI Conversationalists ($n = 55, 20.9\%$):** These students resolve uncertainty by staying inside the chat interface and interrogating the same AI—asking why B is wrong (65.5%), asking the first AI to compare explanations (45.5%), requesting a DOI link from the AI (65.5%), asking the AI to explain step 3→4 (83.6%), and asking why its answer changed (56.4%).

4. **C1 — Cross-AI & Peer Triangulators ($n = 46, 17.5\%$):** Instead of trusting one model, they seek a second opinion from another AI (60.9% in Q_1, 47.8% in Q_2, 65.2% in Q_3, 58.7% in Q_5) or a human classmate (52.2% in Q_4).

5. **C5 — Uncritical AI Conformers ($n = 20, 7.6\%$):** A smaller but educationally critical group that defers to AI authority without verification—abandoning their own answer for D (85.0% in Q_1), using unverified claims while omitting the missing citation (60.0% in Q_3), copying unclear steps (40.0% in Q_4), and accepting sycophantic reversals (45.0% in Q_5).

6. **C6 — AI-Skeptical Anchors ($n = 18, 6.8\%$):** The mirror opposite of C5: when AI conflicts or behaves inconsistently, they reject the AI output outright without further investigation—keeping their own answer B (77.8% in Q_1, 66.7% in Q_5), dropping the unverified claim (61.1% in Q_3), or postponing the task (44.4% in Q_2).

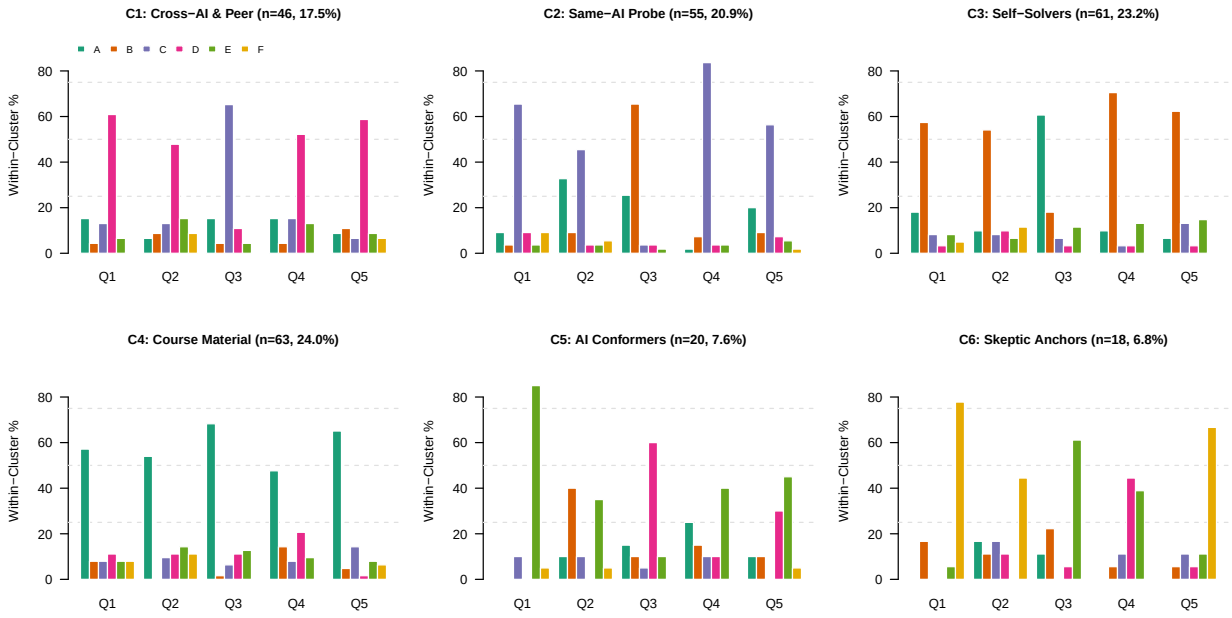


Figure 6: Within-cluster response distributions across Q_1–Q_5 for the six behavioral archetypes ($K = 6$).

4.4 External Validation with Weekly AI Usage (Q_6) and Year of Study (S_4)

Because differences on Q_1–Q_5 are internal to how the clusters were constructed, true validation requires testing whether the clusters differ on external variables withheld from the distance calculation (Q_6 and S_4). Table 3 and Figure 7 present both external validation analyses.

Table 3: External Validation of the $K = 6$ Clusters Against Weekly AI Usage (Q_6) and Year of Study (S_4).

	AI Usage Q_6 (Days/Week)			**Year of Study S_4 (Row % within Cluster, $n = 257$)**				
Cluster ($K = 6$)	**n**	**Mean ± SD**	**Med (IQR)**	**1st Yr**	**2nd Yr**	**3rd Yr**	**4th Yr**	**Grad/Master**
C1: Cross-AI & Peer	46	3.67 ± 2.04	3.0 (2.75)	20.0%	31.1%	15.6%	31.1%	2.2%
C2: Same-AI Probe	55	$\mathbf{4.67 \pm 1.86}$	**5.0** (2.50)	29.6%	14.8%	27.8%	25.9%	1.9%
C3: Self-Solvers	61	3.89 ± 2.29	4.0 (4.00)	19.0%	**37.9%**	10.3%	**32.8%**	0.0%
C4: Course Material	63	4.40 ± 2.01	5.0 (3.00)	23.8%	23.8%	**28.6%**	23.8%	0.0%
C5: AI Conformers	20	$\mathbf{2.85 \pm 1.95}$	**3.0** (3.00)	**42.1%**	5.3%	5.3%	15.8%	**31.6%**
C6: Skeptic Anchors	18	4.44 ± 2.43	5.0 (4.75)	22.2%	33.3%	5.6%	16.7%	**22.2%**

Statistical Tests: Q_6: Kruskal-Wallis $H = 15.023$, df $= 5$, $\mathbf{p = 0.0103}$; One-way ANOVA $F(5, 257) = 3.148$, $\mathbf{p = 0.0089}$.
S_4: Pearson's $\chi^2 = 77.045$, Monte Carlo simulated $\mathbf{p = 0.0001}$ (10,000 replicates). (At $K = 5$: H-test $p = 0.0097$, $\chi^2 = 68.13, p = 0.0001$).

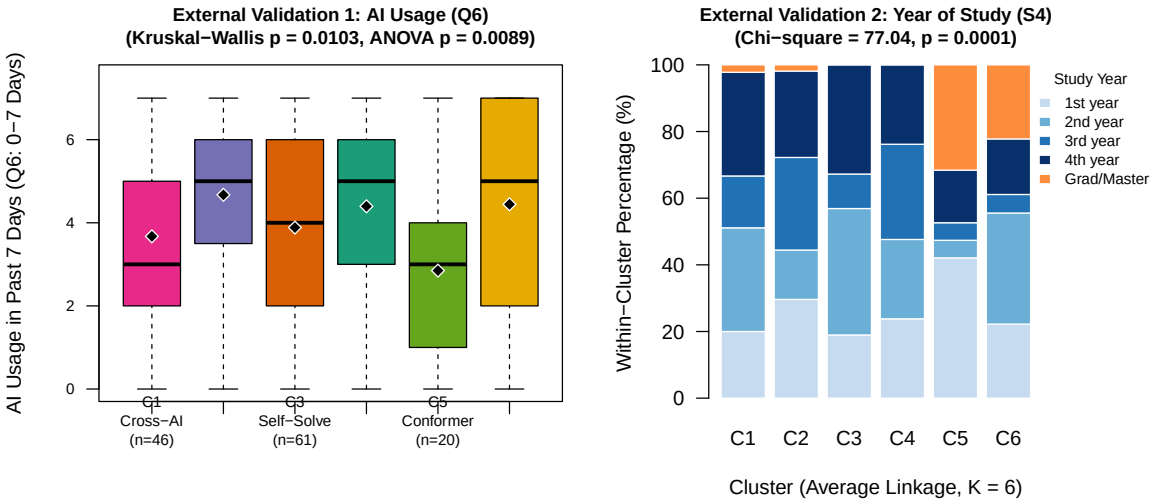


Figure 7: External validation of $K = 6$ clusters against Weekly AI Usage (Q_6, left) and Year of Study (S_4, right).

Both external validation tests are statistically significant and reveal compelling behavioral insights:

- **AI Usage Frequency (Q_6, $p = 0.0103$):** Counter to the naive assumption that heavy AI users blindly trust AI, **Uncritical AI Conformers (C5)** actually have the *lowest* weekly AI usage ($\bar{Q}_6 = 2.85 \pm 1.95$ days/week, median 3.0), whereas **Same-AI Conversationalists (C2)** have the *highest* usage ($\bar{Q}_6 = 4.67 \pm 1.86$ days/week, median 5.0), closely followed by **AI-Skeptical Anchors (C6)** (4.44 days) and **Course-Material Verifiers (C4)** (4.40 days). Frequent users have developed habits of iterative follow-up prompting or healthy skepticism toward AI hallucinations, whereas infrequent users are more prone to treat a single AI response as final authority.
- **Year of Study (S_4, $p = 0.0001$):** Active academic verification (C3 *Independent Problem-Solvers* and C4 *Course-Material Verifiers*) is overwhelmingly concentrated among undergraduate cohorts (Years 1–4 account for 100% of known respondents in C3 and C4). Conversely, **Uncritical AI Conformers (C5)** are disproportionately composed of 1st-year freshmen (42.1%, who may lack domain confidence to challenge an AI explanation) and time-constrained Graduate/Master's students (31.6%).

5 Conclusions and Limitations

Summary of Findings: (1) Answering **RQ1**, VGU students do not respond uniformly to uncertain AI outputs; nearly half (47.1%) engage in rigorous academic verification—either via course materials (C4, 24.0%) or independent mathematical derivation (C3, 23.2%)—while 38.4% rely on secondary AI/peer consultas (C2, 20.9%; C1, 17.5%), and smaller minorities exhibit uncritical conformity (C5, 7.6%) or skeptical rejection (C6, 6.8%). (2) Answering **RQ2**, Average Linkage (UPGMA) strongly outperforms Single and Complete Linkage on low-dimensional nominal Hamming spaces by resolving discrete merge-height ties into 66 distinct rational levels ($r_{\text{coph}} = 0.6275$). (3) Answering **RQ3**, the identified clusters exhibit substantial subsample stability (ARI = 0.522–0.555) and demonstrate strong external validity across both weekly AI usage ($p = 0.0103$) and academic seniority ($p = 0.0001$).

Limitations: Our findings describe self-reported first intentions among $n = 263$ VGU students rather than observed behavioral logs or academic grades. Furthermore, unweighted Hamming distance treats all nominal mismatches within a question equally and is slightly influenced by the higher baseline match probability of 5-option scenarios (Q_3, Q_4) relative to 6-option scenarios (Q_1, Q_2, Q_5).

Future work could incorporate weighted semantic dissimilarities or longitudinal observational tracking.

References

[1] Andersson, C. (2026). *Data Analysis in High Dimensions: Lecture Notes on Cluster Analysis* (pp. 69–126). Vietnamese-German University (VGU).

[2] Gower, J. C. (1971). A general coefficient of similarity and some of its properties. *Biometrics*, 27(4), 857–871.

[3] Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.

[4] Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218.

A Appendix: Reproducible R Code (`code/analysis.R`)

The complete R script used for data cleaning, Gower/Hamming dissimilarity computation, hierarchical clustering, stability testing, external validation, and figure generation is included below.

```r
#!/usr/bin/env Rscript
# ==============================================================================
# Hierarchical Cluster Analysis: "When AI Disagrees With You"
# Course: Data Analysis in High Dimensions (Prof. Dr. C. Andersson, VGU)
# Team: Tran Trong Nhan (10423160), Nguyen Minh Nhat Huy (10423045),
#       Dao Hoang Minh Triet (10423179)
# ==============================================================================

options(stringsAsFactors = FALSE, width = 120)
library(cluster)
library(stats)
library(graphics)
library(grDevices)

# Set reproducible random seed
set.seed(20261111)

# Paths
base_dir <- "/mnt/hdd-1tb/CSE-VGU/2026-2027/Elective-1"
csv_path <- file.path(
  base_dir,
  "When AI Disagrees With You_ How University Students Respond to Uncertain AI Answers (Responses) - Form Responses 1.csv"
)
out_dir  <- file.path(base_dir, "output")
fig_dir  <- file.path(out_dir, "figures")
tab_dir  <- file.path(out_dir, "tables")

dir.create(fig_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(tab_dir, recursive = TRUE, showWarnings = FALSE)

# ------------------------------------------------------------------------------
# Helper: Adjusted Rand Index (Hubert & Arabie 1985)
# ------------------------------------------------------------------------------
adjusted_rand_index <- function(x, y) {
  tab <- table(x, y)
  n <- sum(tab)
  if (n <= 1) return(1.0)
  sum_nij2 <- sum(choose(tab, 2))
  sum_ai2  <- sum(choose(rowSums(tab), 2))
  sum_bj2  <- sum(choose(colSums(tab), 2))
  total_pairs <- choose(n, 2)
  expected_index <- (sum_ai2 * sum_bj2) / total_pairs
  max_index <- 0.5 * (sum_ai2 + sum_bj2)
  denom <- max_index - expected_index
  if (abs(denom) < 1e-12) return(1.0)
  (sum_nij2 - expected_index) / denom
}

# Helper to save plots as both PDF (for LaTeX) and PNG (for Slides)
save_dual_plot <- function(filename_stem, width_in, height_in, plot_fn) {
  pdf_file <- file.path(fig_dir, paste0(filename_stem, ".pdf"))
  png_file <- file.path(fig_dir, paste0(filename_stem, ".png"))
  pdf(pdf_file, width = width_in, height = height_in, useDingbats = FALSE)
  plot_fn()
  dev.off()
  png(png_file, width = width_in * 150, height = height_in * 150, res = 150)
  plot_fn()
  dev.off()
}

# ------------------------------------------------------------------------------
# 1. Load & Preprocess Data
# ------------------------------------------------------------------------------
raw_df <- read.csv(csv_path, check.names = FALSE, encoding = "UTF-8")
colnames(raw_df) <- c(
  "Timestamp", "S1_Enrolled", "S2_Consent", "S3_Major", "S4_Year",
  "Q1_Raw", "Q2_Raw", "Q3_Raw", "Q4_Raw", "Q5_Raw", "Q6_Days"
)

```

```r
# Filter for eligibility (S1 == "Yes" & S2 == "Yes")
df <- raw_df[trimws(raw_df$S1_Enrolled) == "Yes" & trimws(raw_df$S2_Consent) == "Yes", ]
df$ID <- sprintf("R%03d", seq_len(nrow(df)))

# Map Q1-Q5 text responses to option codes A-F / A-E
map_q1 <- function(x) {
  x <- trimws(x)
  ifelse(x == "Check the relevant course material.", "A",
  ifelse(x == "Solve the question again independently, without consulting the AI explanation.", "B",
  ifelse(x == "Ask the same AI why option B is incorrect.", "C",
  ifelse(x %in% c("Ask a different AI the same question again.",
                  "Ask a different AI the original question."), "D",
  ifelse(x == "Change my answer to D without further checking.", "E",
  ifelse(x == "Keep my answer B without further checking.", "F", NA_character_))))))
}

map_q2 <- function(x) {
  x <- trimws(x)
  ifelse(x == "Check the relevant course material.", "A",
  ifelse(x == "Solve the question independently without using either AI explanation.", "B",
  ifelse(x == "Send the second explanation to the first AI and ask it to compare them.", "C",
  ifelse(x == "Ask a third AI the original question.", "D",
  ifelse(x == "Choose the answer from the AI I use more often, without further checking.", "E",
  ifelse(x == "Leave the question unanswered for now and move to another task.", "F",
  NA_character_))))))
}

map_q3 <- function(x) {
  x <- trimws(x)
  ifelse(x == "Independently search for another source to verify the claim.", "A",
  ifelse(grepl("^Ask the same AI for the paper", x), "B",
  ifelse(x == "Ask a different AI whether the claim is correct.", "C",
  ifelse(x == "Use the claim in my work but omit the citation.", "D",
  ifelse(x == "Remove the claim from my work without investigating further.", "E",
  NA_character_)))))
}

map_q4 <- function(x) {
  x <- trimws(x)
  ifelse(x == "Look up the relevant concept or formula in course materials.", "A",
  ifelse(x == "Attempt to derive step 4 from step 3 myself.", "B",
  ifelse(x == "Ask the same AI to explain that specific transition.", "C",
  ifelse(x == "Ask a classmate to explain that specific transition.", "D",
  ifelse(x == "Use the solution without clarifying that step.", "E", NA_character_)))))
}

map_q5 <- function(x) {
  x <- trimws(x)
  ifelse(x == "Check the relevant course material.", "A",
  ifelse(x == "Solve the question independently.", "B",
  ifelse(x == "Ask the AI to compare both answers and explain why it changed.", "C",
  ifelse(x == "Ask a different AI the original question.", "D",
  ifelse(x == "Accept D without further checking.", "E",
  ifelse(x == "Keep B without further checking.", "F", NA_character_))))))
}

df$Q1 <- factor(map_q1(df$Q1_Raw), levels = c("A", "B", "C", "D", "E", "F"))
df$Q2 <- factor(map_q2(df$Q2_Raw), levels = c("A", "B", "C", "D", "E", "F"))
df$Q3 <- factor(map_q3(df$Q3_Raw), levels = c("A", "B", "C", "D", "E"))
df$Q4 <- factor(map_q4(df$Q4_Raw), levels = c("A", "B", "C", "D", "E"))
df$Q5 <- factor(map_q5(df$Q5_Raw), levels = c("A", "B", "C", "D", "E", "F"))

stopifnot(all(colSums(is.na(df[, c("Q1", "Q2", "Q3", "Q4", "Q5")])) == 0))

df$S3_Clean <- ifelse(trimws(df$S3_Major) == "" | trimws(df$S3_Major) == "Prefer not to answer",
                      "Unspecified", trimws(df$S3_Major))
df$S4_Clean <- ifelse(trimws(df$S4_Year) == "" | trimws(df$S4_Year) == "Prefer not to answer",
                      "Unspecified", trimws(df$S4_Year))
df$S4_Clean <- factor(
  df$S4_Clean,
  levels = c("1st year", "2nd year", "3rd year", "4th year", "Graduate / Master's student", "Unspecified")
```

```r
)
df$Q6_Num <- as.integer(trimws(df$Q6_Days))
df$Profile <- paste(df$Q1, df$Q2, df$Q3, df$Q4, df$Q5, sep = "-")

# ------------------------------------------------------------------------------
# 2. Marginal Distributions & Distance Matrix
# ------------------------------------------------------------------------------
q_cols <- c("Q1", "Q2", "Q3", "Q4", "Q5")
marginal_list <- list()
for (q in q_cols) {
  tb <- table(df[[q]])
  pct <- round(100 * prop.table(tb), 1)
  marginal_list[[q]] <- data.frame(
    Question = q,
    Option = names(tb),
    Count = as.integer(tb),
    Percentage = as.numeric(pct)
  )
}
marginal_df <- do.call(rbind, marginal_list)
write.csv(marginal_df, file.path(tab_dir, "marginal_q1_q5.csv"), row.names = FALSE)

df_q <- df[, q_cols]
rownames(df_q) <- df$ID

dist_hamming <- daisy(df_q, metric = "gower")
dist_vec <- as.numeric(dist_hamming)

hc_single   <- hclust(dist_hamming, method = "single")
hc_complete <- hclust(dist_hamming, method = "complete")
hc_average  <- hclust(dist_hamming, method = "average")

coph_single   <- cor(dist_hamming, cophenetic(hc_single))
coph_complete <- cor(dist_hamming, cophenetic(hc_complete))
coph_average  <- cor(dist_hamming, cophenetic(hc_average))

# ------------------------------------------------------------------------------
# 3. Partition Evaluation (K = 2..8) & Stability Analysis
# ------------------------------------------------------------------------------
k_range <- 2:8
eval_rows <- list()
for (m_name in c("Single", "Complete", "Average")) {
  hc_obj <- switch(m_name, Single = hc_single, Complete = hc_complete, Average = hc_average)
  for (k in k_range) {
    cl <- cutree(hc_obj, k = k)
    sil <- silhouette(cl, dist_hamming)
    sil_vals <- sil[, "sil_width"]
    sz <- sort(as.integer(table(cl)), decreasing = TRUE)
    eval_rows[[length(eval_rows) + 1]] <- data.frame(
      Method = m_name,
      K = k,
      Avg_Silhouette = round(mean(sil_vals), 4),
      Neg_Sil_Count = sum(sil_vals < 0),
      Neg_Sil_Pct = round(100 * mean(sil_vals < 0), 1),
      Max_Cluster_Size = max(sz),
      Max_Cluster_Pct = round(100 * max(sz) / nrow(df_q), 1),
      Min_Cluster_Size = min(sz),
      Singletons = sum(sz == 1),
      Cluster_Sizes = paste(sz, collapse = "/")
    )
  }
}
eval_df <- do.call(rbind, eval_rows)
write.csv(eval_df, file.path(tab_dir, "linkage_k_evaluation.csv"), row.names = FALSE)

# Sensitivity & Subsampling Stability (K = 2..6)
set.seed(20261111)
n_perm <- 100
n_sub  <- 100
sub_size <- floor(0.80 * nrow(df_q))

stability_rows <- list()
sub_ari_store <- list()

for (m_name in c("Single", "Complete", "Average")) {
  m_code <- tolower(m_name)
```

```r
  hc_full <- switch(m_name, Single = hc_single, Complete = hc_complete, Average = hc_average)
  for (k in 2:6) {
    cl_full <- cutree(hc_full, k = k)
    names(cl_full) <- rownames(df_q)

    perm_aris <- numeric(n_perm)
    for (b in seq_len(n_perm)) {
      perm_idx <- sample(seq_len(nrow(df_q)))
      df_perm <- df_q[perm_idx, , drop = FALSE]
      d_perm <- daisy(df_perm, metric = "gower")
      hc_perm <- hclust(d_perm, method = m_code)
      cl_perm <- cutree(hc_perm, k = k)
      perm_aris[b] <- adjusted_rand_index(cl_full, cl_perm[names(cl_full)])
    }

    sub_aris <- numeric(n_sub)
    for (b in seq_len(n_sub)) {
      sub_idx <- sort(sample(seq_len(nrow(df_q)), size = sub_size, replace = FALSE))
      df_sub <- df_q[sub_idx, , drop = FALSE]
      d_sub <- daisy(df_sub, metric = "gower")
      hc_sub <- hclust(d_sub, method = m_code)
      cl_sub <- cutree(hc_sub, k = k)
      sub_aris[b] <- adjusted_rand_index(cl_full[rownames(df_sub)], cl_sub)
    }
    if (m_name == "Average") {
      sub_ari_store[[paste0("K=", k)]] <- sub_aris
    }

    stability_rows[[length(stability_rows) + 1]] <- data.frame(
      Method = m_name,
      K = k,
      Perm_ARI_Mean = round(mean(perm_aris), 4),
      Perm_ARI_SD   = round(sd(perm_aris), 4),
      Perm_ARI_Min  = round(min(perm_aris), 4),
      Sub80_ARI_Mean = round(mean(sub_aris), 4),
      Sub80_ARI_SD   = round(sd(sub_aris), 4),
      Sub80_ARI_Median = round(median(sub_aris), 4),
      Sub80_ARI_IQR  = round(IQR(sub_aris), 4)
    )
  }
}
stability_df <- do.call(rbind, stability_rows)
write.csv(stability_df, file.path(tab_dir, "stability_sensitivity_ari.csv"), row.names =
    FALSE)

# ------------------------------------------------------------------------------
# 4. Cluster Assignments & Profiling (Average Linkage K = 5 and K = 6)
# ------------------------------------------------------------------------------
df$Cluster_K5 <- cutree(hc_average, k = 5)
df$Cluster_K6 <- cutree(hc_average, k = 6)

# Descriptive archetype labels for K = 6
archetype_k6 <- c(
  "1" = "C1: Cross-AI & Peer Triangulators",
  "2" = "C2: Same-AI Conversationalists",
  "3" = "C3: Independent Problem-Solvers",
  "4" = "C4: Course-Material Verifiers",
  "5" = "C5: Uncritical AI Conformers",
  "6" = "C6: AI-Skeptical Anchors"
)
df$Archetype_K6 <- archetype_k6[as.character(df$Cluster_K6)]

write.csv(
  df[, c("ID", "S3_Clean", "S4_Clean", "Q1", "Q2", "Q3", "Q4", "Q5", "Q6_Num",
         "Profile", "Cluster_K5", "Cluster_K6", "Archetype_K6")],
  file.path(tab_dir, "cleaned_responses.csv"),
  row.names = FALSE
)

# Export K=6 within-cluster percentage profile table
k6_prof_rows <- list()
for (cl_id in 1:6) {
  sub_df <- df[df$Cluster_K6 == cl_id, ]
  for (q in q_cols) {
    pcts <- round(100 * prop.table(table(factor(sub_df[[q]], levels =
        c("A","B","C","D","E","F")))), 1)
```

```r
    k6_prof_rows[[length(k6_prof_rows) + 1]] <- data.frame(
      Cluster = cl_id,
      Archetype = archetype_k6[as.character(cl_id)],
      n = nrow(sub_df),
      Question = q,
      Opt_A = pcts["A"], Opt_B = pcts["B"], Opt_C = pcts["C"],
      Opt_D = pcts["D"], Opt_E = pcts["E"],
      Opt_F = ifelse(q %in% c("Q3", "Q4"), NA, pcts["F"])
    )
  }
}
k6_prof_df <- do.call(rbind, k6_prof_rows)
write.csv(k6_prof_df, file.path(tab_dir, "k6_cluster_profiles.csv"), row.names = FALSE)

# Export External Validation Tables (Q6 and S4 for K=6 and K=5)
q6_val_k6 <- do.call(data.frame, aggregate(
  df$Q6_Num,
  by = list(Cluster = df$Cluster_K6, Archetype = df$Archetype_K6),
  FUN = function(x) c(n = length(x), Mean = round(mean(x), 2), SD = round(sd(x), 2),
                      Median = median(x), IQR = IQR(x))
))
q6_val_k6 <- q6_val_k6[order(q6_val_k6$Cluster), ]
write.csv(q6_val_k6, file.path(tab_dir, "external_validation_q6_k6.csv"), row.names = FALSE)

known_s4 <- df$S4_Clean != "Unspecified"
s4_ct_k6 <- table(Cluster = df$Cluster_K6[known_s4], Year =
    droplevels(df$S4_Clean[known_s4]))
s4_pct_k6 <- round(100 * prop.table(s4_ct_k6, margin = 1), 1)
write.csv(as.data.frame.matrix(s4_ct_k6), file.path(tab_dir,
    "external_validation_s4_counts_k6.csv"))
write.csv(as.data.frame.matrix(s4_pct_k6), file.path(tab_dir,
    "external_validation_s4_pct_k6.csv"))

# ------------------------------------------------------------------------------
# 5. Generate Publication-Quality Figures
# ------------------------------------------------------------------------------

# Figure 1: Marginal Response Distributions across Q1-Q5
save_dual_plot("fig01_marginal_responses", 8.5, 4.8, function() {
  par(mar = c(4.5, 5.5, 3.2, 8.5), xpd = TRUE)
  mat_pct <- matrix(0, nrow = 6, ncol = 5,
                    dimnames = list(c("A", "B", "C", "D", "E", "F"),
                                    c("Q1: AI Disagrees\nwith You",
                                      "Q2: Two AIs\nDisagree",
                                      "Q3: Missing\nPaper (5 opts)",
                                      "Q4: Unclear\nStep (5 opts)",
                                      "Q5: AI Flips\nAnswer")))
  for (j in seq_along(q_cols)) {
    q <- q_cols[j]
    tb <- prop.table(table(df[[q]])) * 100
    mat_pct[names(tb), j] <- as.numeric(tb)
  }
  cols <- c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e", "#e6ab02")
  bp <- barplot(mat_pct, beside = TRUE, col = cols, ylim = c(0, 48),
                ylab = "Percentage of Respondents (%)",
                main = "Marginal Distribution of Student Responses Across Scenarios Q1-Q5 (n
                    = 263)",
                las = 1, cex.names = 0.88, cex.axis = 0.9, cex.main = 1.05, border = "white")
  abline(h = seq(10, 40, by = 10), col = "gray88", lty = 2)
  barplot(mat_pct, beside = TRUE, col = cols, ylim = c(0, 48), add = TRUE,
          axes = FALSE, axisnames = FALSE, border = "white")
  # Add value labels on top of bars > 0
  for (j in 1:5) {
    for (i in 1:6) {
      if (mat_pct[i, j] > 0) {
        text(bp[i, j], mat_pct[i, j] + 1.5, sprintf("%.0f%%", mat_pct[i, j]), cex = 0.65,
            col = "gray20")
      }
    }
  }
  legend("topright", inset = c(-0.22, 0),
         legend = c("Option A", "Option B", "Option C", "Option D", "Option E", "Option F"),
         fill = cols, border = "white", bty = "n", cex = 0.85, title = "Choice Code")
})

# Figure 2: Pairwise Normalized Hamming Distance Distribution
```

```r
save_dual_plot("fig02_hamming_distribution", 7.5, 4.5, function() {
  par(mar = c(4.5, 4.8, 3.2, 2.0))
  dist_tab <- table(factor(round(dist_vec, 1), levels = c(0, 0.2, 0.4, 0.6, 0.8, 1.0)))
  dist_pct <- 100 * prop.table(dist_tab)
  bp <- barplot(dist_pct, col = "#386cb0", border = "white", ylim = c(0, 45),
                xlab = "Normalized Hamming / Gower Distance d(X_i, X_j)",
                ylab = "Percentage of Respondent Pairs (%)",
                main = "Distribution of Pairwise Normalized Hamming Distances (34,453
                    Pairs)",
                las = 1)
  grid(nx = NA, ny = NULL, col = "gray88", lty = 2)
  barplot(dist_pct, col = "#386cb0", border = "white", ylim = c(0, 45), add = TRUE, axes =
      FALSE)
  text(bp, as.numeric(dist_pct) + 2.2,
       labels = sprintf("%.2f%%\n(%d pairs)", as.numeric(dist_pct), as.integer(dist_tab)),
       cex = 0.78, col = "gray15")
})

# Figure 3: Dendrogram Comparison (Single, Complete, Average Linkage)
save_dual_plot("fig03_dendrograms_comparison", 11.0, 4.2, function() {
  par(mfrow = c(1, 3), mar = c(2.5, 4.0, 3.2, 1.0))
  plot(hc_single, labels = FALSE, hang = -1, col = "#e41a1c",
       main = sprintf("Single Linkage\n(Cophenetic r = %.3f)", coph_single),
       xlab = "", sub = "", ylab = "Normalized Hamming Distance", ylim = c(0, 1.05))
  abline(h = 0.35, lty = 2, col = "gray40")

  plot(hc_complete, labels = FALSE, hang = -1, col = "#ff7f00",
       main = sprintf("Complete Linkage\n(Cophenetic r = %.3f)", coph_complete),
       xlab = "", sub = "", ylab = "Normalized Hamming Distance", ylim = c(0, 1.05))
  rect.hclust(hc_complete, k = 6, border = "gray35")

  plot(hc_average, labels = FALSE, hang = -1, col = "#1f78b4",
       main = sprintf("Average Linkage / UPGMA\n(Cophenetic r = %.3f)", coph_average),
       xlab = "", sub = "", ylab = "Normalized Hamming Distance", ylim = c(0, 1.05))
  rect.hclust(hc_average, k = 6, border = "#d95f02")
})

# Helper to draw a full-view annotated dendrogram with K = 6 boxes
plot_full_dendrogram <- function(hc, title_str, tree_col, cut_line_h,
                                 cluster_ids = NULL, cl_map = NULL) {
  par(mar = c(3.6, 4.6, 4.2, 1.5), xpd = NA)
  plot(hc, labels = FALSE, hang = -1, col = tree_col,
       main = title_str, xlab = "", sub = "",
       ylab = "Normalized Hamming Distance (Merge Height)",
       ylim = c(-0.08, 1.08), las = 1, cex.main = 1.15, cex.lab = 1.0)
  par(xpd = FALSE)
  abline(h = seq(0, 1, by = 0.2), col = "gray88", lty = 3)
  abline(h = cut_line_h, col = "#d95f02", lty = 2, lwd = 1.8)

  box_cols <- c("#e41a1c", "#377eb8", "#4daf4a", "#984ea3", "#ff7f00", "#a65628")
  rects <- rect.hclust(hc, k = 6, border = box_cols)

  # Compute x-centers of each rectangle in dendrogram order
  x_start <- 1
  leg_labels <- character(length(rects))
  has_narrow <- any(sapply(rects, length) < 15)
  par(xpd = NA)
  for (i in seq_along(rects)) {
    idxs <- rects[[i]]
    sz <- length(idxs)
    x_mid <- x_start + (sz - 1) / 2
    if (!is.null(cluster_ids) && !is.null(cl_map)) {
      cid <- as.character(cluster_ids[idxs[1]])
      lbl <- sprintf("%s (n=%d)", cl_map[cid], sz)
      leg_labels[i] <- lbl
    } else {
      lbl <- sprintf("G%d (n=%d)", i, sz)
      leg_labels[i] <- lbl
    }
    if (!has_narrow || sz >= 15) {
      text(x = x_mid, y = -0.025, labels = lbl, col = box_cols[i],
           cex = 0.72, font = 2)
    }
    x_start <- x_start + sz
  }
  if (has_narrow) {
```

```
    legend("topright", inset = c(0.01, 0.06), legend = leg_labels,
           col = box_cols, lwd = 2.2, bty = "o", bg = "white",
           box.col = "gray80", cex = 0.78, ncol = 2, title = "K = 6 Groups (Left to Right)")
  }
  text(x = 263 * 0.99, y = cut_line_h + 0.025,
       labels = sprintf("K = 6 cut (h = %.3f)", cut_line_h),
       adj = c(1, 0), col = "#d95f02", cex = 0.85, font = 2)
  par(xpd = FALSE)
}

# Figure 3a: Full-View Single Linkage Dendrogram
save_dual_plot("fig03a_dendrogram_single", 13.0, 5.2, function() {
  h_cut <- mean(rev(hc_single$height)[5:6])
  plot_full_dendrogram(
    hc_single,
    sprintf("Single Linkage Dendrogram (n = 263, Cophenetic r = %.3f) - Severe Chaining & 5 Singletons", coph_single),
    "#e41a1c", h_cut
  )
})

# Figure 3b: Full-View Complete Linkage Dendrogram
save_dual_plot("fig03b_dendrogram_complete", 13.0, 5.2, function() {
  h_cut <- mean(rev(hc_complete$height)[5:6])
  plot_full_dendrogram(
    hc_complete,
    sprintf("Complete Linkage Dendrogram (n = 263, Cophenetic r = %.3f) - Discrete Grid & Ceiling Ties at h = 1.0", coph_complete),
    "#ff7f00", h_cut
  )
})

# Figure 3c: Full-View Average Linkage (UPGMA) Dendrogram
save_dual_plot("fig03c_dendrogram_average", 13.0, 5.2, function() {
  h_cut <- mean(rev(hc_average$height)[5:6])
  cl_map <- c(
    "1" = "C1: Cross-AI",
    "2" = "C2: Same-AI",
    "3" = "C3: Self-Solve",
    "4" = "C4: Course-Mat",
    "5" = "C5: Conform",
    "6" = "C6: Skeptic"
  )
  plot_full_dendrogram(
    hc_average,
    sprintf("Average Linkage (UPGMA) Dendrogram (n = 263, Cophenetic r = %.3f) - Clear K = 6 Cluster Division", coph_average),
    "#1f78b4", h_cut, cluster_ids = df$Cluster_K6, cl_map = cl_map
  )
})

# Figure 4: Silhouette Width & Largest Cluster % across K = 2..8
save_dual_plot("fig04_silhouette_comparison", 9.0, 4.3, function() {
  par(mfrow = c(1, 2), mar = c(4.5, 4.5, 3.0, 1.5))

  # Left panel: Average Silhouette Width
  sil_s <- eval_df$Avg_Silhouette[eval_df$Method == "Single"]
  sil_c <- eval_df$Avg_Silhouette[eval_df$Method == "Complete"]
  sil_a <- eval_df$Avg_Silhouette[eval_df$Method == "Average"]

  plot(k_range, sil_a, type = "b", pch = 19, lwd = 2.2, col = "#1f78b4",
       ylim = c(-0.27, 0.24), xlab = "Number of Clusters (K)",
       ylab = "Average Silhouette Width",
       main = "Average Silhouette Width vs. K", las = 1)
  grid(col = "gray88", lty = 2)
  lines(k_range, sil_c, type = "b", pch = 17, lwd = 2.0, col = "#ff7f00")
  lines(k_range, sil_s, type = "b", pch = 15, lwd = 2.0, col = "#e41a1c")
  abline(h = 0, lty = 3, col = "gray40")
  points(6, sil_a[k_range == 6], cex = 2.0, col = "#1f78b4", lwd = 2)
  legend("bottomleft", legend = c("Average (UPGMA)", "Complete", "Single"),
         col = c("#1f78b4", "#ff7f00", "#e41a1c"), pch = c(19, 17, 15),
         lwd = 2, bty = "n", cex = 0.85)

  # Right panel: Largest Cluster Share (%) showing Chaining Effect
  max_s <- eval_df$Max_Cluster_Pct[eval_df$Method == "Single"]
  max_c <- eval_df$Max_Cluster_Pct[eval_df$Method == "Complete"]
```

```r
  max_a <- eval_df$Max_Cluster_Pct[eval_df$Method == "Average"]

  plot(k_range, max_a, type = "b", pch = 19, lwd = 2.2, col = "#1f78b4",
       ylim = c(20, 102), xlab = "Number of Clusters (K)",
       ylab = "Largest Cluster Share (% of n = 263)",
       main = "Chaining Diagnostic: Largest Cluster %", las = 1)
  grid(col = "gray88", lty = 2)
  lines(k_range, max_c, type = "b", pch = 17, lwd = 2.0, col = "#ff7f00")
  lines(k_range, max_s, type = "b", pch = 15, lwd = 2.0, col = "#e41a1c")
  legend("bottomleft", legend = c("Average (UPGMA)", "Complete", "Single"),
         col = c("#1f78b4", "#ff7f00", "#e41a1c"), pch = c(19, 17, 15),
         lwd = 2, bty = "n", cex = 0.85)
})

# Figure 5: Subsampling Stability (ARI across 100x 80% Subsamples) & Tie Sensitivity
save_dual_plot("fig05_stability_ari", 8.5, 4.3, function() {
  par(mfrow = c(1, 2), mar = c(4.5, 4.5, 3.0, 1.5))

  # Left panel: Boxplot of 100x 80% Subsampling ARI for Average Linkage (K = 2..6)
  boxplot(sub_ari_store, col = "#a6cee3", border = "#1f78b4",
          ylim = c(0, 1.0), xlab = "Number of Clusters (Average Linkage)",
          ylab = "Adjusted Rand Index (ARI)",
          main = "80% Subsampling Stability (100 Runs)", las = 1)
  grid(nx = NA, ny = NULL, col = "gray88", lty = 2)
  means_sub <- sapply(sub_ari_store, mean)
  points(seq_along(means_sub), means_sub, pch = 18, col = "#e31a1c", cex = 1.4)
  legend("bottomright", legend = c("Median & IQR", "Mean ARI"),
         pch = c(15, 18), col = c("#a6cee3", "#e31a1c"), bty = "n", cex = 0.82)

  # Right panel: Row-Order Permutation Mean ARI across methods (K = 2..6)
  ks <- 2:6
  p_a <- stability_df$Perm_ARI_Mean[stability_df$Method == "Average"]
  p_c <- stability_df$Perm_ARI_Mean[stability_df$Method == "Complete"]
  p_s <- stability_df$Perm_ARI_Mean[stability_df$Method == "Single"]

  plot(ks, p_a, type = "b", pch = 19, lwd = 2.2, col = "#1f78b4",
       ylim = c(0, 1.05), xlab = "Number of Clusters (K)",
       ylab = "Mean Permutation ARI (100 Shuffles)",
       main = "Tie / Row-Order Sensitivity Check", las = 1, xaxt = "n")
  axis(1, at = ks)
  grid(col = "gray88", lty = 2)
  lines(ks, p_c, type = "b", pch = 17, lwd = 2.0, col = "#ff7f00")
  lines(ks, p_s, type = "b", pch = 15, lwd = 2.0, col = "#e41a1c")
  legend("topright", legend = c("Average", "Complete", "Single"),
         col = c("#1f78b4", "#ff7f00", "#e41a1c"), pch = c(19, 17, 15),
         lwd = 2, bty = "n", cex = 0.82)
})

# Figure 6: Behavioral Cluster Profiles (Heatmap of Modal & Option Percentages for K = 6)
save_dual_plot("fig06_cluster_profiles", 10.5, 5.5, function() {
  par(mfrow = c(2, 3), mar = c(4.0, 4.2, 3.0, 1.2))
  cols <- c("#1b9e77", "#d95f02", "#7570b3", "#e7298a", "#66a61e", "#e6ab02")
  short_titles <- c(
    "C1: Cross-AI & Peer (n=46, 17.5%)",
    "C2: Same-AI Probe (n=55, 20.9%)",
    "C3: Self-Solvers (n=61, 23.2%)",
    "C4: Course Material (n=63, 24.0%)",
    "C5: AI Conformers (n=20, 7.6%)",
    "C6: Skeptic Anchors (n=18, 6.8%)"
  )
  for (cl_id in 1:6) {
    sub_df <- df[df$Cluster_K6 == cl_id, ]
    mat_c <- matrix(0, nrow = 6, ncol = 5,
                    dimnames = list(c("A","B","C","D","E","F"), q_cols))
    for (j in seq_along(q_cols)) {
      tb <- 100 * prop.table(table(factor(sub_df[[q_cols[j]]], levels =
          c("A","B","C","D","E","F"))))
      mat_c[, j] <- as.numeric(tb)
    }
    barplot(mat_c, beside = TRUE, col = cols, ylim = c(0, 95),
            main = short_titles[cl_id], ylab = "Within-Cluster %",
            las = 1, border = "white", cex.main = 0.95)
    abline(h = c(25, 50, 75), col = "gray88", lty = 2)
    barplot(mat_c, beside = TRUE, col = cols, ylim = c(0, 95),
            add = TRUE, axes = FALSE, axisnames = FALSE, border = "white")
    if (cl_id == 1) {
```

```
      legend("topleft", legend = c("A", "B", "C", "D", "E", "F"),
             fill = cols, border = "white", horiz = TRUE, bty = "n", cex = 0.78)
    }
  }
})

# Figure 7: External Validation -- AI Usage (Q6) and Year of Study (S4) across K = 6 Clusters
save_dual_plot("fig07_external_validation", 10.0, 4.5, function() {
  par(mfrow = c(1, 2), mar = c(5.5, 4.5, 3.2, 1.5))

  # Left panel: Boxplot + means of Q6 (AI Usage Days) by Cluster
  cl_labels <- c("C1\nCross-AI\n(n=46)", "C2\nSame-AI\n(n=55)",
                 "C3\nSelf-Solve\n(n=61)", "C4\nCourse-Mat\n(n=63)",
                 "C5\nConformer\n(n=20)", "C6\nSkeptic\n(n=18)")
  cl_cols <- c("#e7298a", "#7570b3", "#d95f02", "#1b9e77", "#66a61e", "#e6ab02")
  boxplot(Q6_Num ~ Cluster_K6, data = df, names = cl_labels, col = cl_cols,
          ylab = "AI Usage in Past 7 Days (Q6: 0-7 Days)", xlab = "",
          main = "External Validation 1: AI Usage (Q6)\n(Kruskal-Wallis p = 0.0103, ANOVA p
              = 0.0089)",
          las = 1, cex.axis = 0.78, cex.main = 0.95, ylim = c(0, 7.5))
  q6_means <- tapply(df$Q6_Num, df$Cluster_K6, mean)
  points(1:6, q6_means, pch = 18, col = "white", cex = 1.6)
  points(1:6, q6_means, pch = 18, col = "black", cex = 1.2)

  # Right panel: Stacked bar chart of Year of Study (S4) by Cluster
  par(mar = c(5.5, 4.5, 3.2, 6.8), xpd = TRUE)
  s4_prop <- prop.table(t(s4_ct_k6), margin = 2) * 100
  colnames(s4_prop) <- c("C1", "C2", "C3", "C4", "C5", "C6")
  yr_cols <- c("#c6dbef", "#6baed6", "#2171b5", "#08306b", "#fd8d3c")
  barplot(s4_prop, col = yr_cols, border = "white",
          ylab = "Within-Cluster Percentage (%)",
          xlab = "Cluster (Average Linkage, K = 6)",
          main = "External Validation 2: Year of Study (S4)\n(Chi-square = 77.04, p =
              0.0001)",
          las = 1, cex.main = 0.95)
  legend("topright", inset = c(-0.34, 0),
         legend = c("1st year", "2nd year", "3rd year", "4th year", "Grad/Master"),
         fill = yr_cols, border = "white", bty = "n", cex = 0.78, title = "Study Year")
})

cat("\nAll figures and CSV tables successfully generated in:", out_dir, "\n")
```